

Yichao Jin, Ph.D.

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Specialization: Behavioral AI / Health Economics / Behavioral Economics / Discrete Choice Experiments

Education

University of Texas at Dallas	May 2026
Ph.D. in Public Policy and Political Economy	
- Successfully Defended: April 2026 - Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan” — ProQuest - Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong. - Distinction: Dean of Graduate Education Dissertation Research Award (2025).	
University of Texas at Dallas	May 2026
M.S. in Social Data Analytics and Research	
University of Queensland	2019
Master of Development Economics	
University of California, Riverside	2017
B.A. in Economics	

Research Agenda

My research develops behavioral models and AI-driven simulation frameworks to understand why individuals make suboptimal health decisions, and how policy design can be improved through causally-valid computational inference. I integrate Discrete Choice Experiments with structural econometrics and large language models to build Behavioral Digital Twins—empirically-grounded synthetic populations for counterfactual policy simulation. My work spans health economics, behavioral economics, and AI governance.

Working Papers

1. “Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”
Under Review at Economic Analysis and Policy — [Link to Full Draft](#)
 - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
 - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.
2. “Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework”
Working Paper (Target: J. Biomedical Informatics) — [SSRN Link](#). Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule

($\pi = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{static}}$), the Static-BDT Anchor reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952 . Presented at AIE Summer Conference 2026, DAISY 2026 (ACM CHIL 2026), and AIxPH 2026.

3. **“Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects”**

Working Paper — [SSRN Link](#). Develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination access as a testbed, evaluates nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments.

4. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**

Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **AIxPH 2026 (1st Annual Conference on AI and Public Health, Baltimore, MD)**: Empirical-Frontier Regularization for LLM Synthetic Agents — Poster Presentation.
- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.

- **UT Dallas Graduate Research Symposium 2025:** Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

Dohyeong Kim (Chair)

University of Texas at Dallas

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Robert E. Holmes Jr. Professor of Public Policy,
GIS & Social Data Analytics

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